

EMIL ALVARO SERRANO DANGUILAN

SENIOR FULL-STACK SOFTWARE ENGINEER •
FLUTTER / DART DEVELOPER • AI SYSTEMS
ARCHITECT

Senior Flutter • Full-Stack • Mobile / Backend • Software
Architecture • Technical Leadership • AI Systems



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PROFESSIONAL PROFILE

Hands-on senior software engineer and technical architect with experience across Flutter/Dart mobile development, full-stack web engineering, backend APIs, realtime systems, databases, Linux infrastructure, self-hosted services, and AI/ML product development. Builds complete products from architecture and UX through frontend, backend, data, integrations, deployment and optimization. Combines conventional software engineering with deep specialization in local/edge AI, realtime voice systems, multilingual translation, model orchestration and autonomous agents. Comfortable operating as a senior individual contributor, system architect or technical lead while remaining directly involved in implementation.

ENGINEERING VALUE SNAPSHOT

Flutter + Dart

MOBILE APPLICATION ENGINEERING

Cross-platform application structure, responsive UI, device integration, local storage and realtime workflows.

UI → API → DB

END-TO-END FULL-STACK DELIVERY

React/Vite frontends, Python/Node services, REST/WebSockets, PostgreSQL, Prisma and Firebase.

Edge → GPU

SYSTEMS & INFERENCE ARCHITECTURE

Local mobile runtimes, self-hosted servers and A100/H100-class GPU environments.

40 Public Spaces

EXPERIMENTAL ENGINEERING PORTFOLIO

Public Hugging Face experimentation across agents, speech, multimodal and image-generation systems.

PROFESSIONAL EXPERIENCE

Chief Technology Officer / Full-Stack & AI Systems Architect Current

Eburon AI

Technical ownership spanning product architecture, mobile/web engineering, backend services, realtime voice systems, local/self-hosted AI infrastructure and rapid product R&D.

- Designed complete product architectures spanning Flutter mobile applications, web frontends, Python/Node backend services, databases, realtime communication and AI runtimes.
 - Built and integrated local/self-hosted inference pipelines across mobile, desktop and GPU-server environments, selecting deployment paths based on latency, memory, model size and product constraints.
 - Developed realtime voice architectures combining STT, LLM/intent processing, orchestration, task state, execution and TTS.
 - Designed mobile-first applications for minimal-cloud or offline/local-first operation, including local model runtimes and device-oriented workflows.
 - Worked across Flutter, Dart, React, TypeScript, Python, REST APIs, WebSockets, Firebase, PostgreSQL/Prisma, Linux/VPS infrastructure, Docker-based workflows, reverse proxies and GPU compute.
 - Designed autonomous task orchestration linking assistants to mobile/computer execution agents with normalized task states and verification workflows.
 - Led technical trade-offs around edge-vs-server inference, concurrency, infrastructure requirements, performance and production deployment.
 - Delivered rapid R&D prototypes and demos for internal, customer and investor-facing product validation.
- Emil Alvaro Serrano DangUILAN • Curriculum Vitae

CORE PROFILE

FULL-STACK

Frontend, backend, APIs, databases, realtime state, integration and deployment.

FLUTTER / DART

Mobile-first application architecture, responsive UI, local storage, device workflows and backend integration.

BACKEND SYSTEMS

Python services, Node.js, REST, WebSockets, realtime task/data flows and service integration.

DATA

PostgreSQL, Prisma, Firebase Realtime Database and local device storage.

ARCHITECTURE

System decomposition, orchestration, local-first design, tool calling, task execution and verification.

INFRASTRUCTURE

Linux, Ubuntu, VPS, Docker workflows, reverse proxies, ngrok, background services and GPU compute.

TECHNICAL STACK

LANGUAGES

Dart Python JavaScript
TypeScript HTML CSS SQL

FRAMEWORKS / UI

Flutter React Vite Android

BACKEND / DATA

Node.js REST APIs
WebSockets PostgreSQL
Prisma Firebase

AI / ML

LLMs RAG Agentic AI
Multimodal GGUF ONNX

RUNTIME / INFRA

Ollama llama.cpp
whisper.cpp Linux Docker
VPS

BEST-FIT ROLES

SENIOR FLUTTER / MOBILE

Flutter/Dart app delivery with realtime, local-first and device-integrated systems.

SENIOR FULL-STACK ENGINEER

Owns product slices from frontend through API, data and deployment.

SOFTWARE / SYSTEMS ARCHITECT

Designs complete technical systems and resolves cross-layer engineering trade-offs.

CTO / TECHNICAL LEAD

Hands-on product architecture, R&D, infrastructure and technical strategy.

AI SYSTEMS ENGINEER

Speech, local inference, model orchestration and agentic execution systems.

FULL-STACK & MOBILE ENGINEERING

Flutter / Dart Application Engineering

Builds mobile-first applications with reusable UI structures, local persistence, realtime updates, backend connectivity, native/device interaction and offline/local-first workflows.

Frontend Engineering

Develops responsive application interfaces with React, Vite, JavaScript/TypeScript, HTML and CSS, integrating realtime backend state and API-driven product flows.

Backend APIs & Realtime Systems

Creates Python and Node.js services using REST APIs and WebSockets, including realtime task state, orchestration, integration layers and local/server communication.

Data, Infrastructure & Deployment

Works with PostgreSQL, Prisma, Firebase RTDB, local device storage, Linux/Ubuntu VPS deployments, Docker-based workflows, reverse proxies, ngrok and GPU infrastructure.

SELECTED ENGINEERING PROJECTS

Beatrice — Voice-First Assistant & Orchestration Platform MOBILE + BACKEND + AI

Architected a voice-first system that converts natural-language requests into executable mobile/computer workflows. The design connects microphone input, STT, intent/LLM processing, task normalization, realtime job state, device execution, verification and voice response. Built around a clear separation between orchestration and execution so the assistant can delegate actions and report verified outcomes.

Engineering: Flutter/mobile integration, realtime task state, Firebase RTDB, backend/executor bridge, local/self-hosted models, STT/TTS, device-control workflows.

Eburon Dual Translator REALTIME SPEECH

Designed a two-way multilingual speech translation system targeting offline/local-first mobile operation. Pipeline includes streaming speech recognition, language detection, translation/refinement, target-language validation, streaming TTS and speaker/microphone turn management to prevent TTS feedback into recognition.

Engineering: Flutter/mobile, streaming STT, LLM translation, local inference, Sherpa-ONNX/Whisper-class runtimes, TTS, low-latency state machine design.

Eburon Hub — Local AI Runtime & Server FLUTTER + LOCAL RUNTIME

Designed a mobile-oriented environment that combines local LLM, STT, TTS and image-generation capabilities with user-loadable models and an OpenAI-compatible local server. Explored how Flutter can coordinate native model runtimes while preserving an offline-first product experience.

Engineering: Flutter, GGUF model loading, llama.cpp, whisper.cpp, Sherpa-ONNX, local API server, mobile inference optimization.

Eburon CodeBox — AI-Assisted Development Environment DESKTOP FULL-STACK

Designed an engineering environment using Electron, React/Vite and backend services with coding-agent workflows, terminal integration, browser/computer-use agents, multi-model routing, provider failover, autonomous task execution and job monitoring.

Engineering: Electron, React, Vite, backend services, terminal/process integration, agent orchestration, multi-provider routing and local/remote execution.

Mobile Offline AI Research EDGE SYSTEMS

Conducted R&D into running LLM, STT, TTS and multimodal workloads on constrained mobile hardware. Focus areas include quantized inference, GGUF/ONNX runtimes, mobile Linux/Termux environments, streaming inference, memory constraints and exposing local models through API services.

Engineering: Android/mobile, Termux/Linux, quantized models, GGUF, ONNX, llama.cpp, whisper.cpp, local servers and resource optimization.

HOW I BUILD PRODUCTS

01 • DEFINE Constraints

Clarify user flow, platform limits, offline boundaries & latency.

02 • ARCHITECT System Design

Map UI, services, data, realtime state, APIs & topology.

03 • BUILD Implementation

Implement frontend/mobile, backend services & runtimes.

04 • INTEGRATE End-to-End Flow

Connect components, define contracts & async workflows.

05 • DEPLOY Infrastructure

Package across local devices, VPS/Linux & GPU clusters.

06 • OPTIMIZE Performance

Tune latency, memory, models & service execution flow.

ENGINEERING STRENGTHS

• END-TO-END OWNERSHIP

Can move from product requirement to UI, service logic, data, integration and deployment without losing system-level context.

• MOBILE-FIRST THINKING

Designs around device constraints, local storage, background workflows, realtime interaction and minimal-cloud operation.

• REALTIME ARCHITECTURE

Experienced with WebSockets, realtime databases, streaming speech, normalized task state and low-latency interaction patterns.

• SYSTEMS INTEGRATION

Connects UI, APIs, databases, local runtimes, device-control layers and external services into complete workflows.

• RAPID R&D

Turns research repos, emerging runtimes, models and APIs into functioning prototypes and product architecture quickly.

• TECHNICAL DECISIONS

Balances performance, memory, latency, infrastructure cost, model size, platform limits and product UX.

PUBLIC ENGINEERING FOOTPRINT

Public profiles show hands-on experimentation, model packaging and maintained engineering work. These are presented as supporting evidence of technical depth, while the primary positioning remains full-stack software engineering, Flutter/Dart mobile development and systems architecture.

Hugging Face — MasterDee

PUBLIC PORTFOLIO

Public profile under Emil Alvaro Serrano Danguilan with 40 Spaces spanning agents, speech, image generation, multimodal experiments and model demonstrations. Useful as evidence of continuous hands-on technical exploration and prototyping.

Profile: huggingface.co/MasterDee

GPH-Emilo Hypermodal LLM

OLLAMA

Public Ollama project from Aitek PH Software. The project documentation identifies Emil Alvaro Serrano Danguilan as the software engineer who spearheaded the model initiative, combining software-engineering focus with multimodal capability and Filipino cultural context.

Public model: ollama.com/chatgph/gph-main

Medix-PH

OLLAMA

Public 6.74B Q4_0 model package oriented toward Philippine healthcare communication. The public model page explicitly credits Emil Alvaro Serrano Danguilan as developer.

Public model: ollama.com/chatgph/medix-ph

Eburon Model Family

MODEL PLATFORM

Public Eburon model metadata identifies Emil as the lead developer and maintainer of the Eburon Model Family, covering a broader family of general-purpose, reasoning, coding and vision-oriented models.

Evidence: Public Eburon model metadata on Ollama.

PLATFORM & INFRASTRUCTURE DEPTH

Linux / VPS / Services

Ubuntu/Linux environments, persistent services, background processes, reverse proxies, ngrok tunnels and deployment-oriented debugging.

GPU & Self-Hosted Compute

Architectural experience spanning local CPU/mobile inference through A100/H100-class GPU infrastructure for larger models and concurrent workloads.

Local-First & Offline Systems

Designs for reduced cloud dependency with local model runtimes, on-device storage, local APIs and explicit server/edge boundaries.

Realtime Product Architecture

Streaming speech, task-state synchronization, WebSockets/realtime databases, low-latency execution loops and voice/mobile interaction.

WHAT I BRING TO AN ENGINEERING TEAM

- **Software-first breadth:** Equally comfortable discussing UI behavior, Dart/Flutter code, API boundaries, data flow, Linux services and deployment topology.
- **Hands-on architecture:** Architecture decisions remain grounded in implementation constraints rather than diagram-only design.
- **Cross-domain problem solving:** Can integrate mobile, web, backend, realtime, device-control and AI components where a product requires them.
- **Local/edge specialization:** Understands why cloud-first assumptions fail on privacy, latency, connectivity, cost or device autonomy requirements.
- **Technical leadership:** Translates product goals into engineering systems, evaluates trade-offs and can stay close to code and infrastructure during delivery.
- **Rapid prototyping:** Capable of testing emerging runtimes, repositories and technologies quickly before committing to production architecture.

AI & SPEECH DEPTH

• LLM SYSTEMS

Model orchestration, RAG, tool calling, quantization, local inference and agentic workflows.

• SPEECH

Realtime STT, TTS, speech translation, streaming inference and turn-taking design.

• MULTIMODAL

Vision/multimodal model integration and image-generation workflows.

• LOCAL / EDGE AI

GGUF, ONNX, llama.cpp, Ollama, whisper.cpp and mobile/local inference trade-offs.

MODELS & RUNTIMES

[Ollama](#)[llama.cpp](#)[Gemma](#)

[Qwen](#)[TranslateGemma](#)

[Whisper](#)[whisper.cpp](#)

[Sherpa-ONNX](#)[Supertonic](#)

[Kokoro](#)[Piper](#)[Chatterbox](#)

PORTFOLIO LINKS

GITHUB

[emilalvaroserrano-collab](#)

LINKEDIN

[emil-alvaro-danguilan](#)

HUGGING FACE

huggingface.co/MasterDee

OLLAMA — GPH-EMILO

[chatgph/gph-main](https://ollama.com/chatgph/gph-main)

OLLAMA — MEDIX-PH

[chatgph/medix-ph](https://ollama.com/chatgph/medix-ph)

ENVIRONMENT & TOOLING

[macOS](#)[Linux](#)[Ubuntu](#)

[Android](#)[Git](#)[GitHub](#)[Docker](#)

[VPS](#)[ngrok](#)[A100 / H100](#)

ADDITIONAL INFO

• LANGUAGES

English • Filipino / Tagalog

• WORK PREFERENCE

Remote • International • Project-Based • Full-Time

• AVAILABILITY

Open to senior engineering, architecture and leadership roles.

• REFERENCES

Available upon request